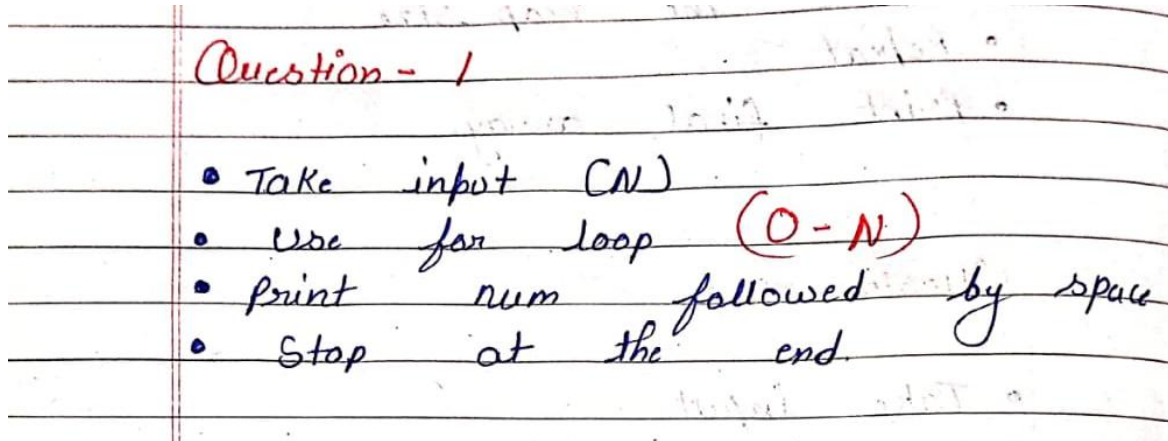


Week 4 Module

Question 01



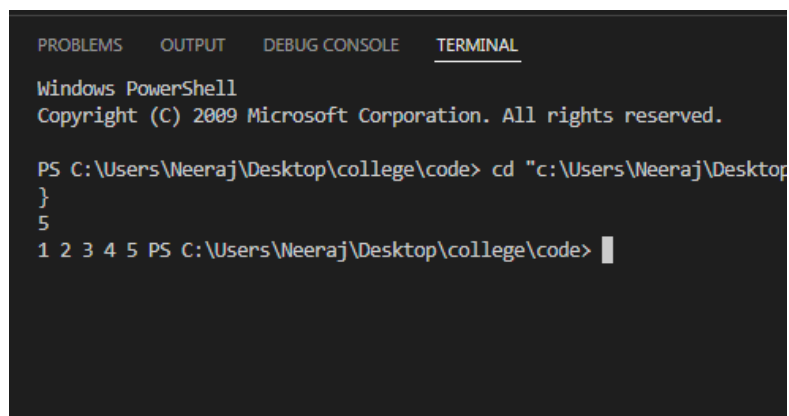
```
import java.util.Scanner;

public class print_num {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        for (int i = 1; i <= n; i++) {
            System.out.print(i + " ");
        }
    }
}
```



Question 02

Question - 2

- Take input (N)
- make variable $[sum = 0;]$
- use for loop $[0-N]$
- Add each num to sum
- print sum.

```
import java.util.Scanner;

public class sum_of {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        long sum = 0;

        for (int i = 1; i <= n; i++) {
            sum += i;
        }

        System.out.println(sum);
    }
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```
tempCodeRunnerFile.java:2: error: class, interface, enum, or record expected
for (int i = 1; i <= n; i++) {
    ^
```

3 errors

PS C:\Users\Neeraj\Desktop\college\code> cd "c:\Users\Neeraj\Desktop\college\code"

5

15

PS C:\Users\Neeraj\Desktop\college\code> |

Question 03

Question - 3

- Take input A, B
- make variable $[result = 1;]$
- use for loop B time.
- multiply result by A
- print result.

```
import java.util.Scanner;

public class power {
    public static void main(String[] args) {

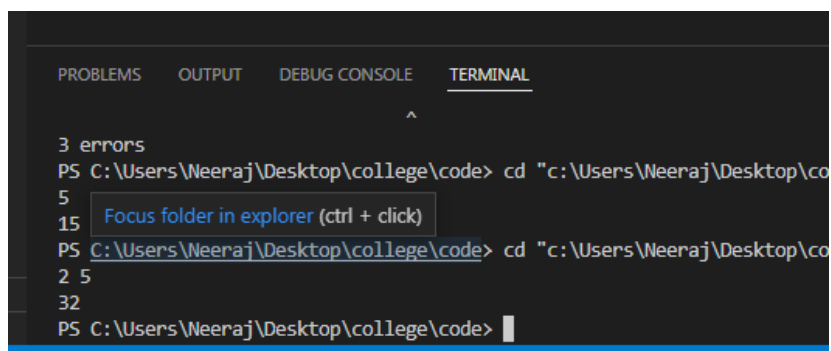
        Scanner sc = new Scanner(System.in);

        int a = sc.nextInt();
        int b = sc.nextInt();

        long result = 1;

        for (int i = 1; i <= b; i++) {
            result *= a;
        }

        System.out.println(result);
    }
}
```



The screenshot shows a terminal window with the following content:

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL
^
3 errors
PS C:\Users\Weeraj\Desktop\college\code> cd "c:\Users\Weeraj\Desktop\co
5
15 Focus folder in explorer (ctrl + click)
PS C:\Users\Weeraj\Desktop\college\code> cd "c:\Users\Weeraj\Desktop\co
2 5
32
PS C:\Users\Weeraj\Desktop\college\code> |
```

Complexity Analysis :-

Question	Time Complexity	Space Complexity
Print 1 to N	$O(N)$	$O(1)$
Sum of N Numbers	$O(N)$	$O(1)$
A^B Using Loop	$O(B)$	$O(1)$